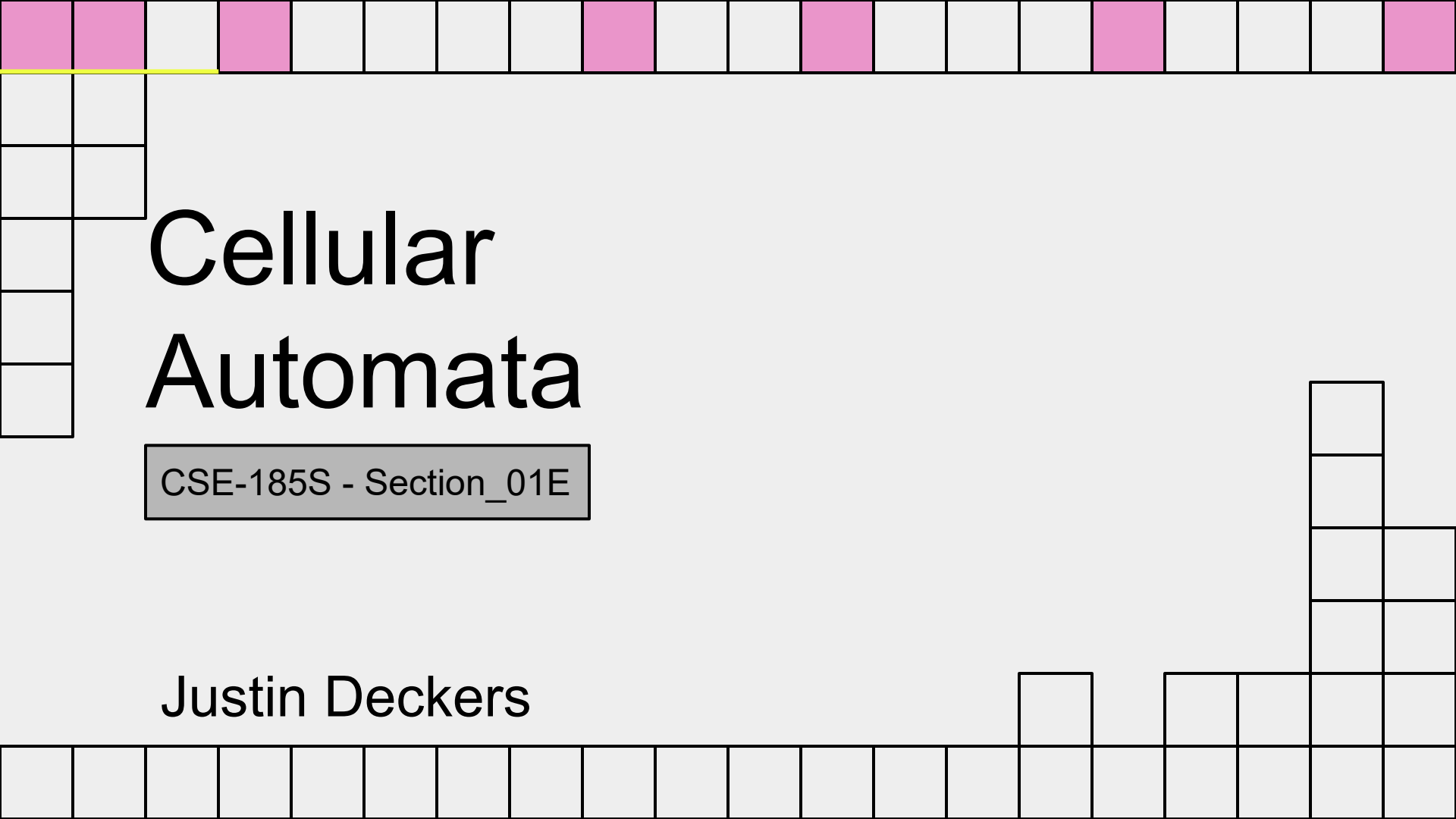




Cellular Automata

CSE-185S - Section_01E

Justin Deckers



What is Cellular Automata?

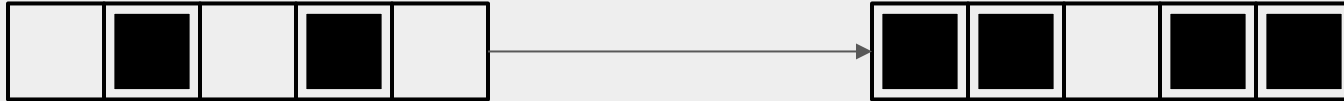
- ☐ Definition: A cellular automaton (CA) is a discrete model studied in computer science, mathematics, and complex systems.
- ☐ Basic Concept: Grid of cells, each in one of a finite number of states. Cells evolve over discrete time steps according to a set of rules.





1 Dimensional CAs

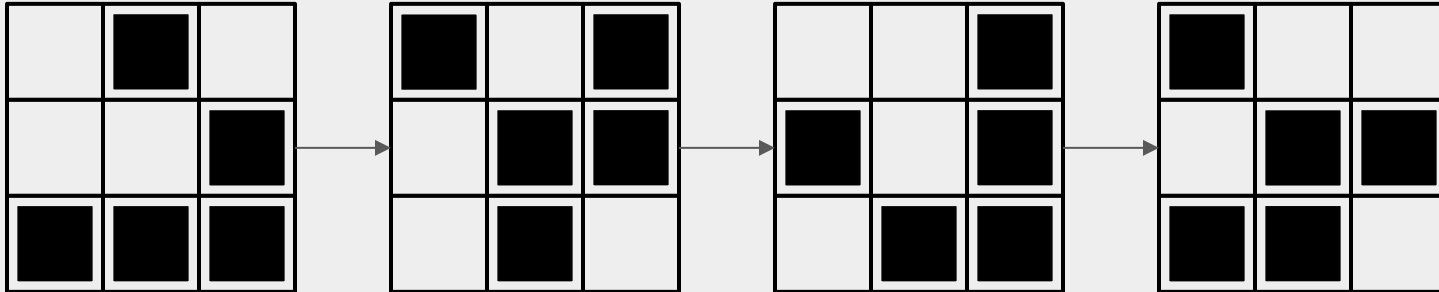
Structure: A single line of cells, each depending on its neighbors (left and right).





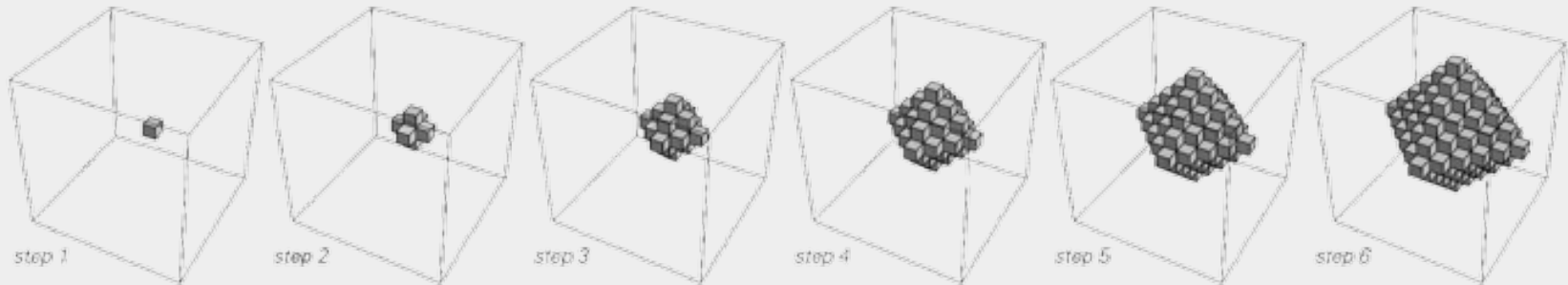
2 Dimensional CAs

Structure: A grid where each cell's next state depends on itself and its neighbors in the four or eight directions



3 Dimensional CAs

Structure: A 3D grid or "cube" of cells, with each cell depending on its neighbors in all 26 possible directions (including up and down).





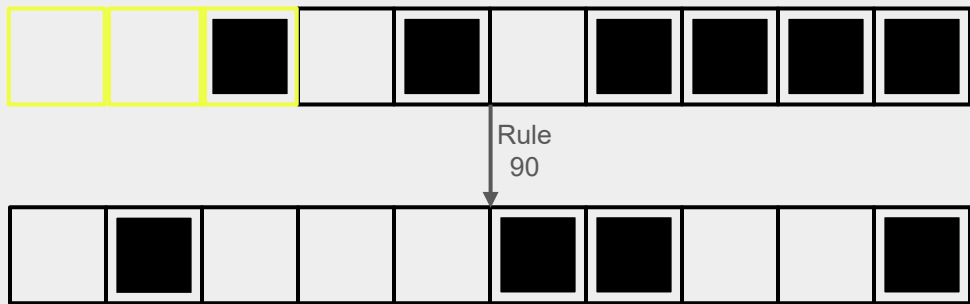
Why Should We Care About ECAs?

- ☐ Applications in Science and Computation
- ☐ Insights into Complexity and Chaos
- ☐ Relevance to Modern Computation



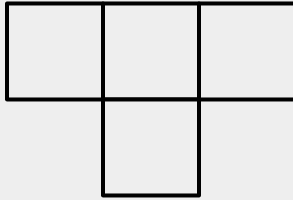
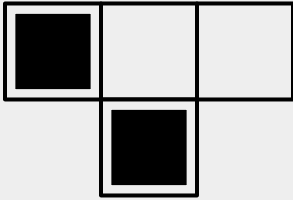
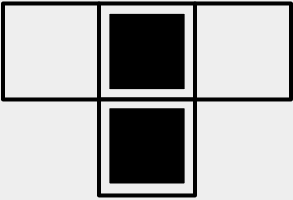
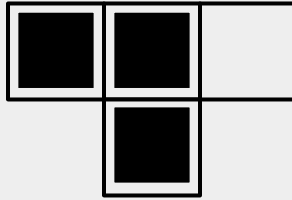
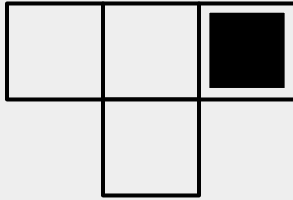
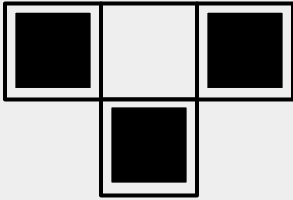
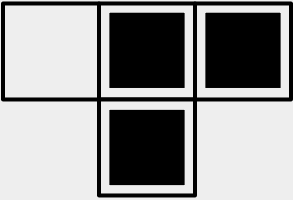
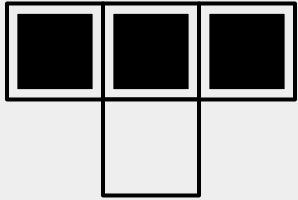


Structure of an ECA





Rule Application



: CA Rule 110





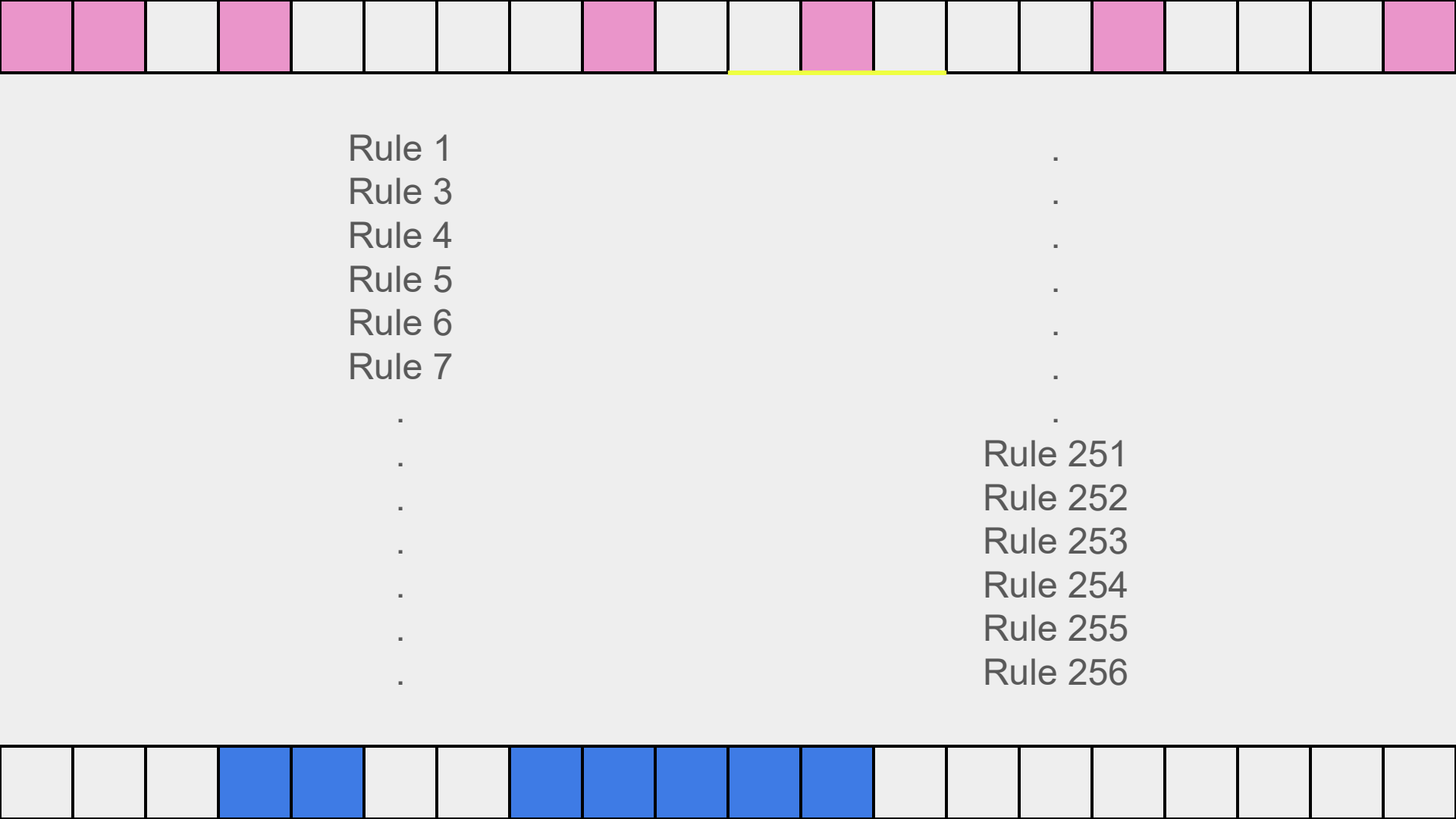
How Many Rules?

With only 2 states & only 8 possible neighbor permutations

$$2^8$$

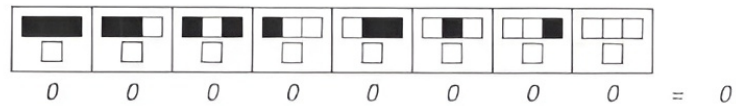
= 256 Total rules



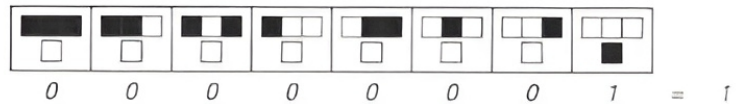




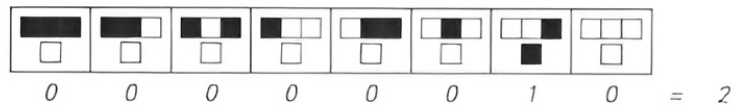
Rule 1:



Rule 2:

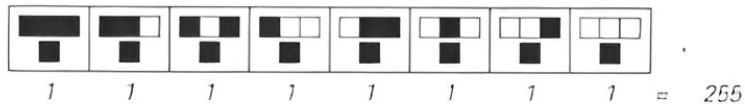


Rule 3:



⋮

Rule 256:





Computer Science

Urban Planning

Physics & Biology

Computation

- ☐ Not just simulations and animations
 - ☐ its ideas power real-world research and applications.
- ☐ Used as a tool for understanding complexity in systems
 - ☐ where simple rules create intricate patterns.



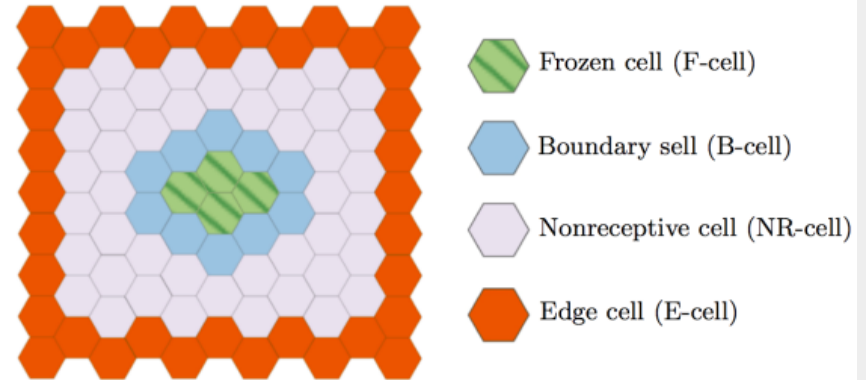
Snowflake Formation

- ☐ Demonstrates pattern formation.
- ☐ Different rules produce unique snowflake patterns.
- ☐ Small changes $\rightarrow$ vastly different fractal structures!



Rule-Based Growth

- ☐ Ice forms when a frozen cell is adjacent.
- ☐ Random diffusion of water particles.
- ☐ Symmetry emerges naturally.





Key Takeaways

- ☐ More than just a graphical representation—it is a conceptual framework driving research in various disciplines.
- ☐ Simple interactions modeled in CA scale up to explain complex, real-world behaviors.





References

- ☐ <https://mathworld.wolfram.com/ElementaryCellularAutomaton.html>
- ☐ <https://www.techtarget.com/searchenterprisedesktop/definition/cellular-automaton>
- ☐ <https://youtube.com/playlist?list=PLRqwX-V7Uu6YrWXvEQFOGbCt6cX83Xunm&si=z9UOUtZgtrsQyXWh>
- ☐ <https://plato.stanford.edu/entries/cellular-automata/supplement.html>
- ☐ https://itp.uni-frankfurt.de/~gros/StudentProjects/Projects_2020/projekt_yan_huck/





**THANK YOU
FOR BEING
HERE TODAY!**



ANY QUESTIONS?